

AI IMPACT ON JOBS: LAYOFF RISK ANALYSIS

A Data Analytics Project Using Python, PostgreSQL & Power BI

Prepared by: Vishal Yadav

B.Tech (Electronics & Telecommunication Engineering), Final Year

Shree L. R. Tiwari College of Engineering, Mumbai

Data Analyst / Data Science Candidate Portfolio Project

Dataset Source: Kaggle — “AI Impact on Jobs and Layoff Risk Dataset”

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1. Executive Summary

This report documents an end-to-end data analytics project examining how artificial intelligence adoption and task automation relate to employee layoff risk. The project was built to demonstrate practical proficiency across the three tools most commonly required in Data Analyst and Data Scientist roles: Python for exploratory analysis and predictive modeling, PostgreSQL for structured business querying, and Power BI for interactive dashboard reporting.

Using a 20,000-row employee-level dataset sourced from Kaggle, the analysis identifies which combination of role, industry, and individual factors is most associated with high layoff risk, builds a binary classification model to flag high-risk employees, and packages the findings into a three-page interactive dashboard intended for business stakeholders.

1.1 Key Findings at a Glance

20,000 Employee records analyzed	92.4% Out-of-sample model accuracy	48.2% Highest-risk industry (Manufacturing)	0.3% → 77% Risk range across routine-task levels
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1.2 Headline Insights

- Routine, repetitive, low-creativity job roles carry dramatically higher layoff risk than roles requiring judgment and human interaction — the single strongest pattern in the data.
- Entry-level employees face roughly four times the layoff risk of Senior-level employees (44.5% vs. 11.5%), suggesting reskilling investment should be prioritized at the entry tier.
- Manufacturing, Logistics, and Retail are the highest-risk industries; Education and Healthcare are the lowest-risk, consistent with the routine-task content typical of each sector.
- A Random Forest classifier, validated using 5-fold cross-validation to avoid data leakage, achieves 92.4% out-of-sample accuracy in flagging high-risk employees.
- An engineered feature — the ratio of automated tasks to creativity and human-interaction demands — proved more predictive than any single raw column in the dataset.

1.3 Scope and Honesty Note

Important caveat, stated upfront:

The target variable in this dataset separates very cleanly along automation-related features, which suggests the dataset was generated using an explicit rule rather than sampled from real organizational outcomes. This report treats the modeling results as a demonstration of correct analytical methodology — proper validation, leakage avoidance, and feature engineering — rather than as a claim of real-world predictive validity. This distinction is discussed in full in Section 8.

2. Problem Statement

2.1 Background

As organizations across industries adopt AI tools and automation technologies at increasing scale, the nature of work is shifting. Roles built around routine, repeatable, rule-based tasks face rising displacement risk, while roles that depend on creativity, judgment, and human interaction remain comparatively insulated. Understanding which factors most strongly predict this risk — at both the individual and organizational level — has practical value for workforce planning, career decision-making, and policy design.

2.2 Research Question

This project was designed to answer a specific, concrete question:

Core question:

Which combination of role, industry, and individual factors predicts layoff risk in an AI-disrupted workforce — and can that risk be modeled and flagged reliably using standard data analytics and machine learning techniques?

2.3 Why This Analysis Matters

2.3.1 For HR and Workforce Planning Teams

Identifying which job functions and departments carry the highest automation-driven risk allows HR and workforce planning teams to prioritize reskilling and upskilling investment where it will have the greatest protective effect, rather than spreading limited training budgets evenly across an organization.

2.3.2 For Individual Job Seekers and Employees

Understanding which roles and skill profiles carry higher long-term automation exposure can inform career decisions — for example, which adjacent skills to develop, or which industries may offer more durable employment over the medium term.

2.3.3 For This Project's Author

From a portfolio standpoint, this project was deliberately scoped to demonstrate an end-to-end analytics workflow spanning three tools — Python, SQL, and Power BI — working together on a single coherent business question, rather than three disconnected exercises.

3. Dataset Overview

3.1 Source and Structure

The dataset used in this project is the “AI Impact on Jobs and Layoff Risk Dataset,” sourced from Kaggle. It contains 20,000 employee-level records across 16 original columns, with zero missing values and zero duplicate rows — a clean starting point that required minimal data-cleaning effort, allowing the bulk of the project's time to be spent on analysis, feature engineering, and modeling rather than data wrangling.

20,000 Employee records	16 Original columns	0 Missing values / duplicates	8 / 24 Industries / Job roles
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3.2 Column Reference

Column	Type	Description
Age	Numeric	Employee age in years
Education_Level	Categorical (ordinal)	High School, Bachelor's, Master's, PhD
Years_of_Experience	Numeric	Total years of professional experience
Industry	Categorical (nominal)	8 sectors: Manufacturing, Logistics, Retail, Finance, Telecom, IT, Healthcare, Education
Job_Role	Categorical (nominal)	24 specific roles, ~3 per industry
Company_Size	Categorical (ordinal)	Small, Medium, Large
Job_Level	Categorical (ordinal)	Entry, Mid, Senior
Routine_Task_Percentage	Numeric (0–100)	Share of role's work that is routine / repeatable
Creativity_Requirement	Numeric (0–100)	Degree to which the role demands creative judgment
Human_Interaction_Level	Numeric (0–100)	Degree of interpersonal interaction the role requires
AI_Adoption_Level	Categorical (ordinal)	Low, Medium, High — organization's AI adoption maturity
Number_of_AI_Tools_Used	Numeric	Count of distinct AI tools used by the employee
AI_Usage_Hours_Per_Week	Numeric	Hours per week spent using AI tools
Tasks_Automated_Percentage	Numeric (0–100)	Share of the role's tasks currently automated
AI_Training_Hours	Numeric	Hours of AI-related training received
Layoff_Risk	Categorical (target)	High / Medium / Low — the original 3-class target

3.3 Target Variable Distribution

The original target, `Layoff_Risk`, is a 3-class categorical variable that is nearly perfectly balanced: approximately 33% of records fall into each of High, Medium, and Low risk. For this project, the target was collapsed into a binary variable, `High_Risk_Flag` (High vs. Not High), to support a cleaner business framing — “flag employees at high risk” — and a more interview-friendly classification problem. The resulting binary split is approximately 66% Not High Risk and 34% High Risk, a mild imbalance that was explicitly accounted for during modeling (see Section 6).

3.4 A Note on Data Authenticity

Known limitation, disclosed upfront:

During exploratory analysis (Section 5), it became clear that `Layoff_Risk` separates very cleanly along automation-related features such as `Routine_Task_Percentage` — far more cleanly than would typically be expected from real-world, organically collected workforce data. This pattern is consistent with a synthetically generated dataset built from an explicit underlying rule, rather than one sampled from genuine employment outcomes. This does not reduce the value of the project as a demonstration of correct analytical method — EDA, feature engineering, validation, and dashboarding — but it does mean the model's predictive performance should not be read as evidence of real-world forecasting accuracy. This distinction is revisited explicitly in Section 8 (Limitations).

4. Tools and Methodology

This project was deliberately structured as a three-stage pipeline, with each tool handling the part of the workflow it is best suited for. Python performs data cleaning, exploratory analysis, feature engineering, and predictive modeling; PostgreSQL performs structured business-style querying on the cleaned dataset; and Power BI assembles the results — including the model's output — into an interactive, stakeholder-facing dashboard.

4.1 Technology Stack

Tool	Role in Project	Key Libraries / Features Used
Python	EDA, feature engineering, machine learning	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
PostgreSQL	Business-style structured query analysis	Window functions (RANK, NTILE), CTEs, GROUP BY aggregation
Power BI	Interactive dashboard reporting	DAX measures, multi-page report, integrated ML output

4.2 End-to-End Workflow

The overall pipeline follows a consistent, linear-then-branching structure:

1. Raw CSV (20,000 rows) is loaded and inspected in Python for missing values, duplicates, and data types.
2. Python performs cleaning, exploratory analysis, feature engineering, and trains two classification models.
3. The cleaned dataset is exported in two forms: one for direct import into PostgreSQL, and one (including the model's out-of-sample predictions) for Power BI.
4. PostgreSQL runs 14 business-analysis queries covering aggregation, CTEs, and window functions.
5. Power BI combines descriptive aggregates with the model's predicted risk scores into a three-page interactive report.

4.3 Why This Structure

Rather than treating Python, SQL, and Power BI as three separate, disconnected exercises, the project was designed so that Python's output directly feeds both downstream tools. This mirrors how analytics work is typically structured in industry settings, where a data science or analytics team produces a model, and that model's output is then surfaced to business stakeholders through both ad-hoc SQL queries and a recurring BI dashboard — rather than the model living in a notebook in isolation.

5. Exploratory Data Analysis

Before any modeling was attempted, the dataset was explored to understand how the target variable relates to each feature group: numeric task-characteristic features, categorical role and industry features, and AI-exposure features. This section presents the key findings, supported by the actual charts generated from the dataset.

5.1 Numeric Features by Risk Level

The boxplots below compare the distribution of each numeric feature between the High-Risk and Not-High-Risk groups. The separation is visually striking for task-related features (Routine_Task_Percentage, Tasks_Automated_Percentage, Creativity_Requirement), and far less pronounced for purely personal features (Age, Years_of_Experience) — an early signal that role characteristics, not individual demographics, drive most of the risk signal.

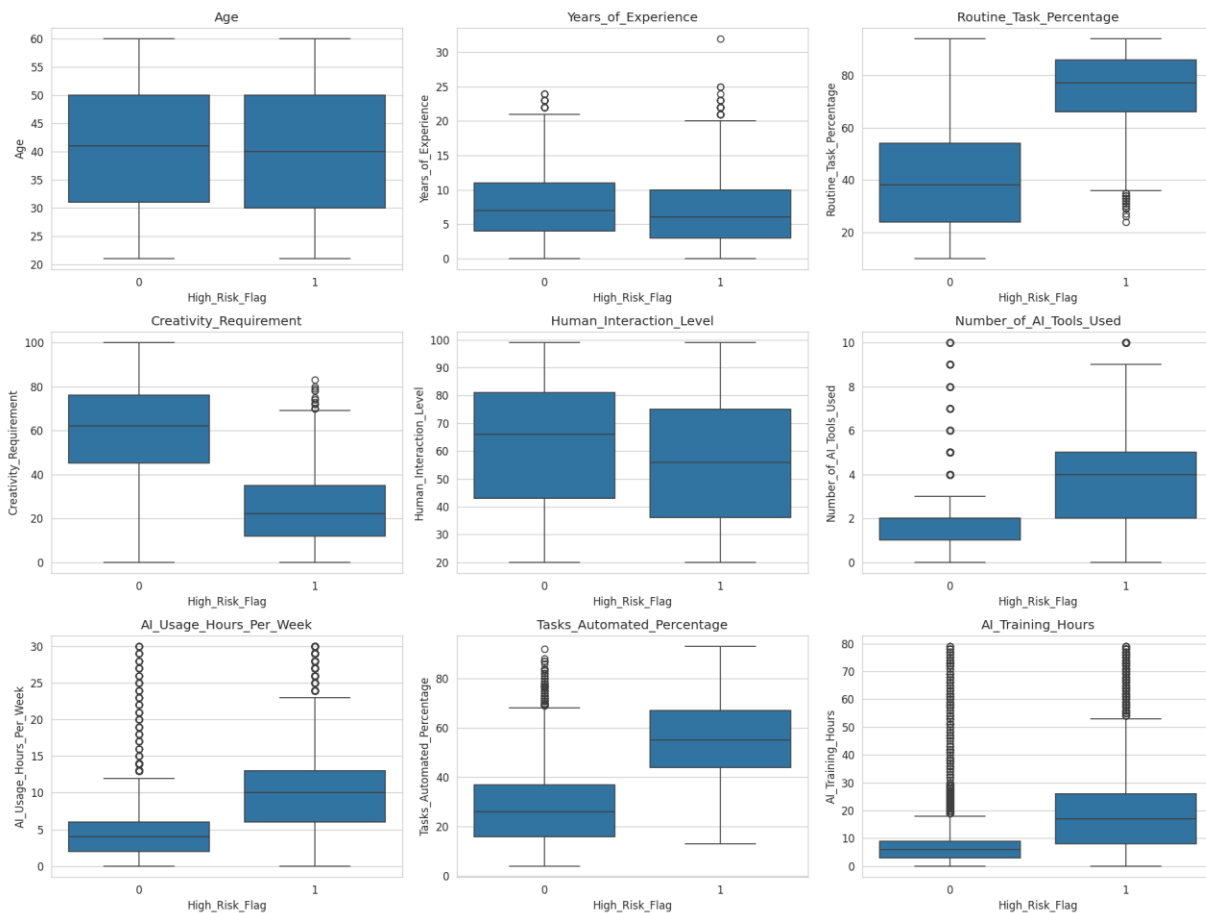


Figure 5.1 — Distribution of numeric features, split by High_Risk_Flag (0 = Not High Risk, 1 = High Risk).

5.2 Categorical Breakdown of Risk

The four panels below show high-risk rate broken down by Industry, AI Adoption Level, Job Level, and Education Level respectively.

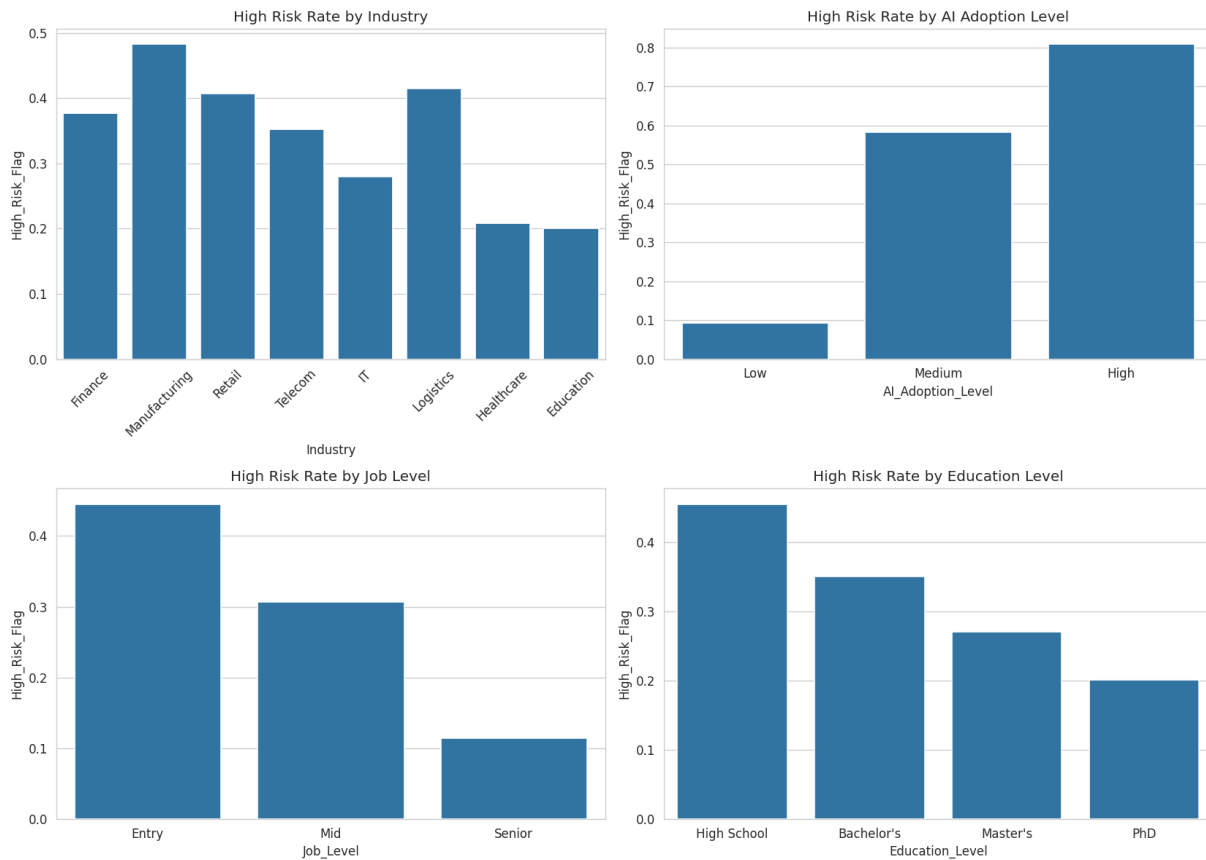


Figure 5.2 — High-risk rate by Industry, AI Adoption Level, Job Level, and Education Level.

5.2.1 Industry

Industry	High-Risk Rate
Manufacturing	48.2%
Logistics	41.5%
Retail	40.7%
Finance	37.7%
Telecom	35.2%
IT	28.0%
Healthcare	20.8%
Education	20.1%

Manufacturing carries the highest risk, consistent with the high proportion of routine, physically repeatable tasks typical of production-line roles. Education and Healthcare — sectors with high human-interaction and judgment requirements — sit at the bottom of the risk ranking.

5.2.2 Job Level

Job Level	High-Risk Rate
Entry	44.5%
Mid	30.7%
Senior	11.5%

Risk decreases sharply with seniority. Entry-level employees face roughly four times the risk of Senior-level employees, a pattern consistent with senior roles typically carrying more judgment-based, less routine responsibilities.

5.2.3 AI Adoption Level

AI Adoption Level	High-Risk Rate	Avg. Tasks Automated %
Low	9.2%	≈ 14%
Medium	58.2%	≈ 47%
High	80.8%	≈ 71%

Organizations or roles with high AI adoption show a more than eight-fold increase in high-risk rate compared to low-adoption roles — the largest gap of any single categorical feature in the dataset.

5.3 Correlation Structure

The correlation heatmap below shows pairwise relationships between all numeric features and the binary target. Routine_Task_Percentage and Tasks_Automated_Percentage both show strong positive correlation with High_Risk_Flag, while Creativity_Requirement and Human_Interaction_Level show negative correlation — confirming the directional intuition that routine, low-creativity, low-interaction roles are most exposed.

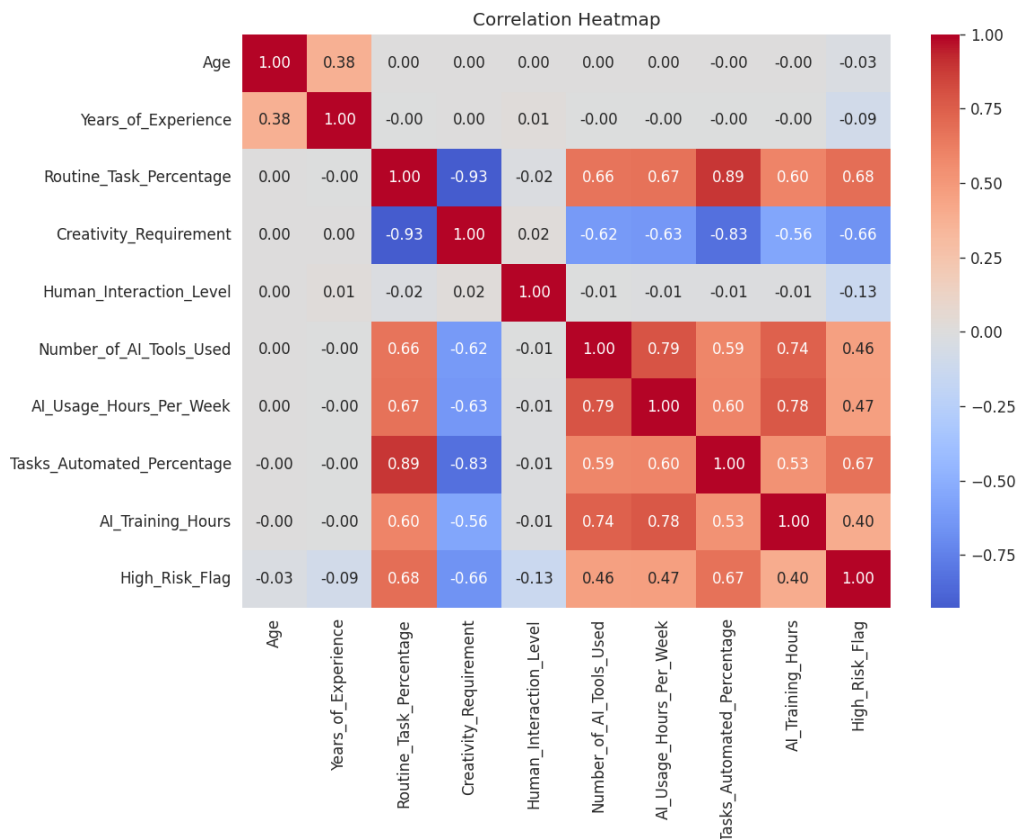


Figure 5.3 — Correlation heatmap of numeric features and the binary target.

5.4 Routine Task Exposure: The Strongest Single Signal

Routine Task Bucket	High-Risk Rate
Low (0–33%)	0.3%
Medium (34–66%)	21.8%
High (67–100%)	77.1%

Key EDA finding:

Routine_Task_Percentage alone behaves as a near-perfect risk gate: employees in low-routine roles are almost never classified as high risk (0.3%), while employees in high-routine roles are high risk more than three-quarters of the time (77.1%). This single feature carries more standalone predictive signal than any other column in the dataset, and motivated the engineered Automation_Exposure_Ratio feature discussed in Section 6.

5.5 Summary of EDA Findings

- Risk concentrates heavily in roles with high routine-task content, low creativity demand, and heavy automation exposure.

- This pattern holds consistently across every categorical cut examined: industry, job level, education level, and AI adoption level.
- Personal demographic features (age, years of experience) show comparatively weak standalone relationships with risk — role characteristics dominate.
- The cleanliness of these relationships foreshadows the data-authenticity caveat discussed in Sections 3.4 and 8.

6. Feature Engineering

Before modeling, the cleaned dataset was transformed into a model-ready feature set. Four distinct engineering decisions were made, each with a specific rationale.

6.1 Binary Target Construction

The original Layoff_Risk column has three classes (High, Medium, Low), nearly balanced at approximately 33% each. This was collapsed into a binary target, High_Risk_Flag, defined as 1 when Layoff_Risk equals “High” and 0 otherwise. This produces a cleaner business framing — “flag employees at high risk” — and results in a mildly imbalanced binary split of approximately 66% / 34%, which is explicitly accounted for in the modeling stage (Section 7).

6.2 Ordinal Encoding

Four categorical features have a natural rank order and were encoded accordingly, rather than one-hot encoded, to preserve that ordinal information for the model:

Feature	Encoding
Education_Level	High School = 0, Bachelor's = 1, Master's = 2, PhD = 3
Job_Level	Entry = 0, Mid = 1, Senior = 2
Company_Size	Small = 0, Medium = 1, Large = 2
AI_Adoption_Level	Low = 0, Medium = 1, High = 2

6.3 One-Hot Encoding

Industry and Job_Role have no natural ordering between categories, so both were one-hot encoded (with the first category dropped in each case to avoid multicollinearity), producing a set of binary indicator columns for each industry and job role.

6.4 Engineered Feature: Automation Exposure Ratio

The single most consequential feature engineering decision in this project was the construction of a new ratio feature, designed to capture how much a role's automatable workload outweighs its human-skill demands:

Automation_Exposure_Ratio =

$\text{Tasks_Automated_Percentage} \div (\text{Creativity_Requirement} + \text{Human_Interaction_Level} + 1)$ The “+ 1” denominator term prevents division by zero for roles with minimal creativity and human-interaction requirements.

This single engineered feature turned out to be the most important predictor in the final Random Forest model (see Section 7.5), outperforming every raw column in the dataset — demonstrating the value of domain-informed feature construction over relying on raw inputs alone.

6.5 Final Feature Set

After encoding, one-hot expansion, and the addition of the engineered ratio feature, the model-ready dataset retains the original 20,000 rows with an expanded column set covering: all original numeric features, four ordinally encoded features, one-hot encoded Industry and Job_Role indicators, the Automation_Exposure_Ratio, and the binary target High_Risk_Flag. String-valued original columns and the original 3-class target were dropped from the modeling dataset, though they are retained in a separate, readable version of the data used for the SQL and Power BI stages.

7. Predictive Modeling

With the engineered feature set complete, two classification models were trained to predict `High_Risk_Flag`: a Logistic Regression baseline and a Random Forest ensemble. Both were evaluated on the same stratified train/test split, and both account for the mild class imbalance identified in Section 6.1.

7.1 Train/Test Split

The data was split 80/20 into training and test sets using a fixed random seed (`random_state = 42`) for reproducibility, with stratification on the target variable to preserve the 66/34 class ratio in both splits.

7.2 Model 1: Logistic Regression (Baseline)

Logistic Regression was chosen as an interpretable baseline. All features were standardized using `StandardScaler` prior to fitting, since Logistic Regression is sensitive to feature scale and unscaled inputs can cause solver convergence issues. The `class_weight` parameter was set to `'balanced'` to compensate for the mild class imbalance, and `max_iter` was set to 1000.

7.3 Model 2: Random Forest (Ensemble)

A Random Forest classifier with 200 trees and a maximum depth of 10 was trained as the primary model, also using `class_weight = 'balanced'`. Random Forest was selected both for its typically strong performance on tabular data with mixed categorical/numeric features, and because it provides a natural mechanism for feature importance ranking (Section 7.5).

7.4 Model Performance Comparison

Model	ROC-AUC	Notes
Logistic Regression	0.996	Scaled features, balanced class weights, interpretable baseline
Random Forest	0.981	200 trees, depth 10, used for final predictions and feature importance

Both models achieve very high ROC-AUC scores. As discussed in Sections 3.4 and 8, this reflects the unusually clean separability of the target variable in this particular dataset, and should be interpreted as a successful demonstration of methodology rather than a claim about real-world forecasting power.

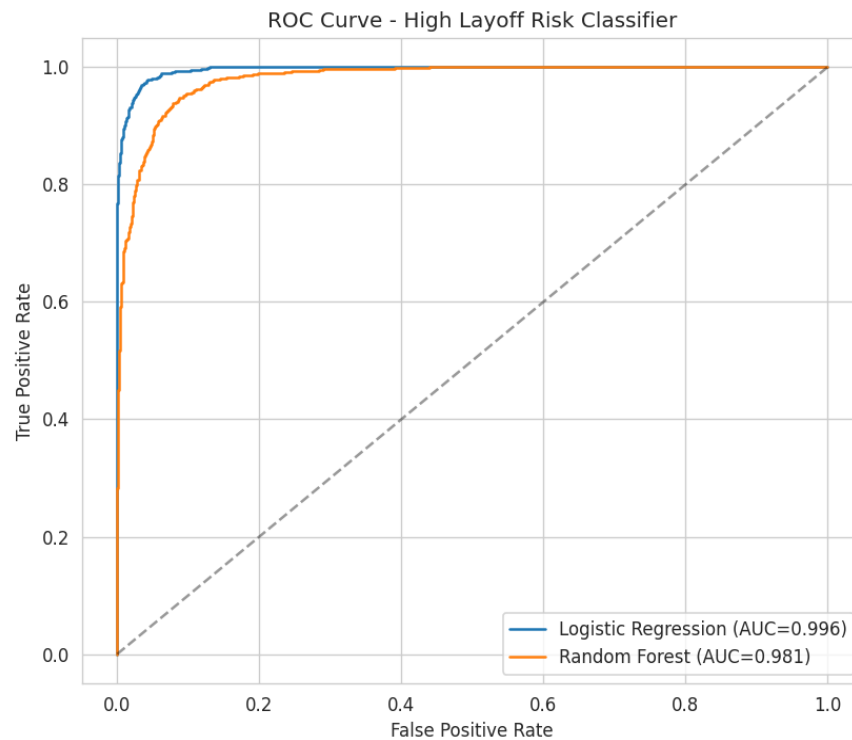


Figure 7.1 — ROC curves for both models on the held-out test set.

7.5 Confusion Matrices

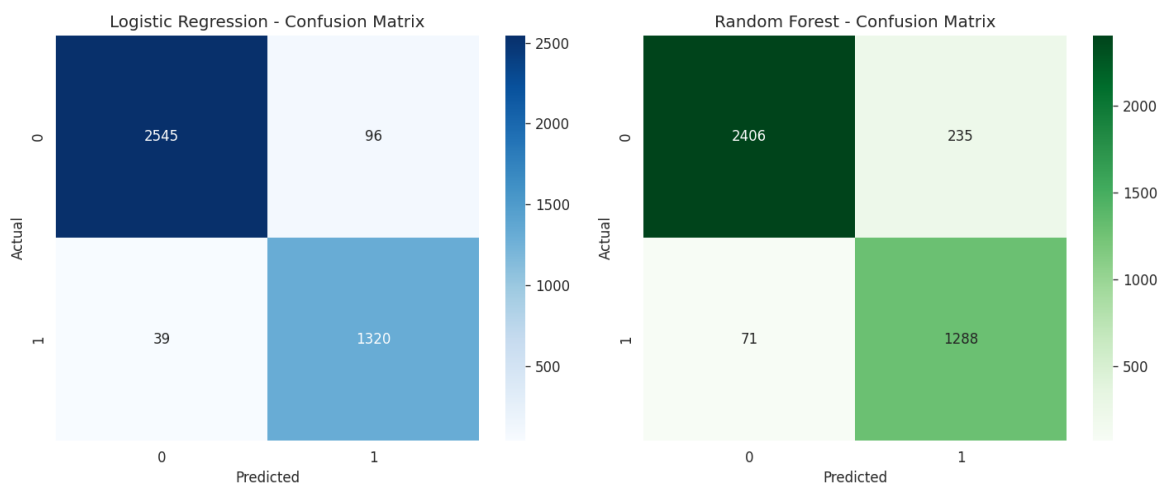


Figure 7.2 — Confusion matrices for Logistic Regression (left) and Random Forest (right) on the test set.

7.6 Model Validation: Avoiding Data Leakage

This subsection documents what is, in the author's assessment, the single most important methodology decision in the entire project, and the one most worth discussing in a technical interview.

7.6.1 The Problem with the Naive Approach

The naive (incorrect) approach:

Generate predictions by running the trained model on the same data it was trained on, and report that as the model's accuracy. This produces an artificially inflated, untrustworthy metric, because the model has already seen those exact rows during training — it is not being tested on anything genuinely unseen. This pattern is known as data leakage, and is one of the most common pitfalls in applied machine learning, particularly among less experienced practitioners.

7.6.2 The Approach Used in This Project

What was done instead:

Per-employee risk scores intended for the Power BI dashboard (Section 9) were generated using scikit-learn's `cross_val_predict` function with 5-fold cross-validation. This ensures that every single row's prediction comes from a model fold in which that row was held out during training — in other words, every prediction is a genuine out-of-sample prediction, even though predictions are ultimately generated for all 20,000 rows.

7.6.3 Result

92.4% Out-of-sample accuracy (5-fold CV)	5 Cross-validation folds used
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This 92.4% figure is the number presented as the project's headline accuracy metric throughout this report and the accompanying dashboard — not the inflated, same-data prediction figure that a less careful analysis might have reported. Making this distinction explicit, and being able to explain why it matters, was treated as a deliverable in its own right.

7.7 Feature Importance

The Random Forest model's built-in feature importance scores reveal which inputs the model relied on most heavily when distinguishing high-risk from non-high-risk employees.

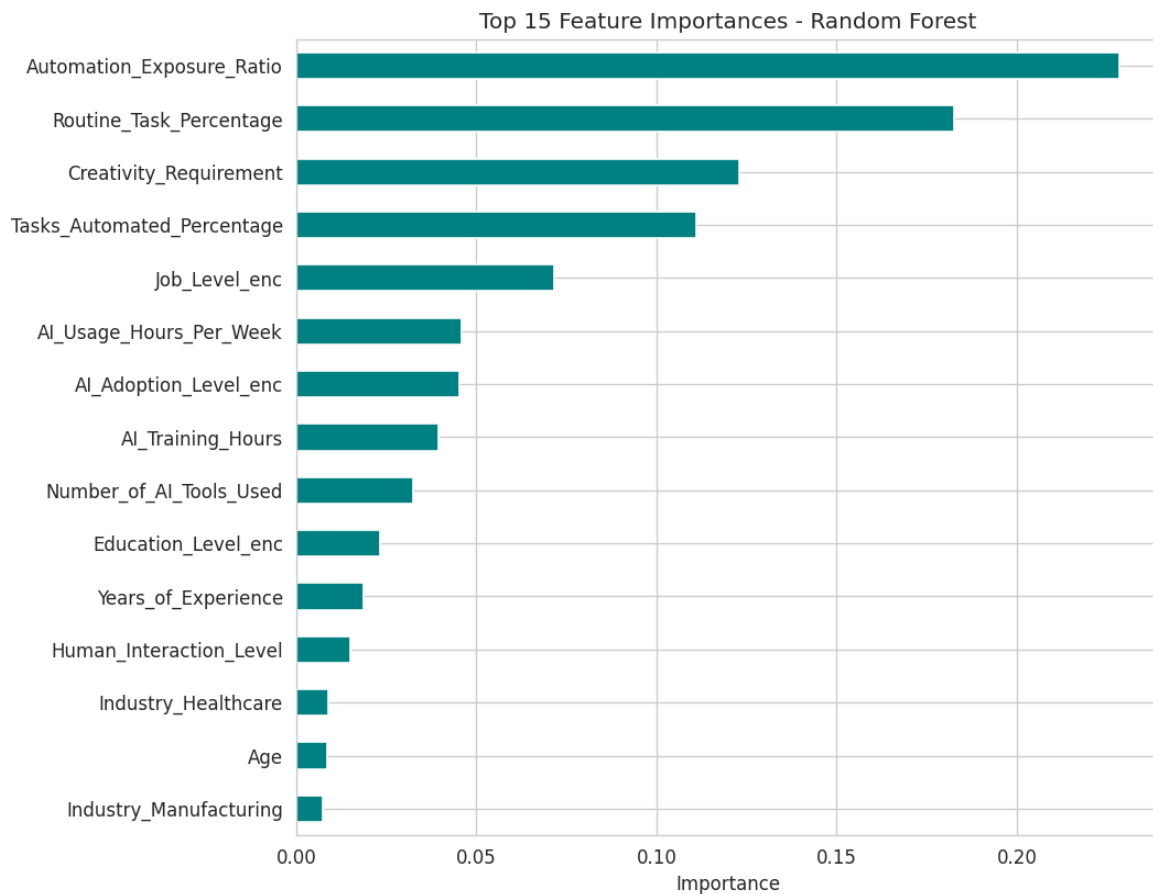


Figure 7.3 — Top 15 feature importances from the Random Forest model.

7.7.1 Top 5 Predictors

Rank	Feature	Interpretation
1	Automation_Exposure_Ratio	Engineered feature; outperformed every raw column
2	Routine_Task_Percentage	Near-perfect risk gate on its own (see Section 5.4)
3	Creativity_Requirement	Inversely related to risk
4	Tasks_Automated_Percentage	Direct automation exposure measure
5	Job_Level (encoded)	Seniority offers measurable protection

The dominance of the engineered Automation_Exposure_Ratio over every individual raw feature is the clearest evidence in this project that thoughtful, domain-informed feature construction adds real value beyond what is available in the raw data alone.

8. SQL Analysis (PostgreSQL)

The cleaned dataset was loaded into a PostgreSQL table, `ai_job_risk`, with the binary `High_Risk_Flag` implemented as a `GENERATED` column derived directly from `layoff_risk` — meaning the database itself computes the flag on every insert, rather than requiring it to be pre-computed in Python. Fourteen business-style analysis queries were then written against this table, covering simple aggregation, common table expressions (CTEs), and window functions.

8.1 Table Schema

```
CREATE TABLE ai_job_risk (
    employee_id          SERIAL PRIMARY KEY,
    age, education_level, years_experience,
    industry, job_role, company_size, job_level,
    routine_task_pct, creativity_requirement,
    human_interaction_level, ai_adoption_level,
    num_ai_tools_used, ai_usage_hrs_per_week,
    tasks_automated_pct, ai_training_hours,
    layoff_risk,
    high_risk_flag INT GENERATED ALWAYS AS
        (CASE WHEN layoff_risk = 'High' THEN 1 ELSE 0 END) STORED
);
```

8.2 Representative Queries

8.2.1 Industry-Wise Risk Ranking (Aggregation)

Total employees, high-risk count, and high-risk percentage per industry, sorted descending.

```
SELECT industry, COUNT(*) AS total_employees,
       SUM(high_risk_flag) AS high_risk_count,
       ROUND(100.0*SUM(high_risk_flag)/COUNT(*),2) AS high_risk_pct
FROM ai_job_risk
GROUP BY industry
ORDER BY high_risk_pct DESC;
```

8.2.2 Top 3 Riskiest Roles per Industry (Window Function — RANK)

`RANK()` partitioned by industry surfaces the highest-risk job roles within each sector.

```
WITH role_risk AS (
    SELECT industry, job_role, COUNT(*) AS total_employees,
           ROUND(100.0*SUM(high_risk_flag)/COUNT(*),2) AS high_risk_pct
    FROM ai_job_risk GROUP BY industry, job_role
),
ranked AS (
    SELECT *, RANK() OVER (
        PARTITION BY industry ORDER BY high_risk_pct DESC
    ) AS risk_rank
    FROM role_risk
```

```
)  
SELECT * FROM ranked WHERE risk_rank <= 3  
ORDER BY industry, risk_rank;
```

8.2.3 Risk by Automation Quartile (Window Function — NTILE)

NTILE(4) splits employees into automation-exposure quartiles to test the risk gradient directly in SQL.

```
WITH percentile_data AS (  
    SELECT employee_id, tasks_automated_pct, high_risk_flag,  
           NTILE(4) OVER (ORDER BY tasks_automated_pct) AS automation_quartile  
    FROM ai_job_risk  
)  
SELECT automation_quartile, COUNT(*) AS total_employees,  
       ROUND(100.0*SUM(high_risk_flag)/COUNT(*),2) AS high_risk_pct  
FROM percentile_data  
GROUP BY automation_quartile  
ORDER BY automation_quartile;
```

8.3 Full Query Coverage

Beyond the three queries detailed above, the full SQL script (14 queries total) also includes:

- CTE-based risk-tier segmentation using CASE expressions to bucket employees by routine-task exposure.
- HAVING-filtered aggregation to identify the top 10 highest-risk job roles, excluding small, statistically noisy groups.
- A window-function query comparing each job role's risk rate to its industry average (deviation analysis).
- Experience-band segmentation testing whether seniority protects against risk independent of job level.
- A comprehensive multi-metric industry summary table, designed to feed directly into Power BI as a pre-aggregated source.

9. Power BI Dashboard

The final stage of the project assembles the SQL aggregates and the Python model's out-of-sample predictions into a three-page interactive Power BI report, designed to be the primary artifact a non-technical stakeholder would actually look at.

9.1 Page 1 — Overview

- KPI cards: Total Employees, High Risk %, Average Predicted Risk Score, Model Accuracy.
- Donut chart showing the original 3-class Layoff_Risk split.
- Horizontal bar chart of High Risk % by Industry.
- Slicers for Industry, Job Level, and Education Level.

9.2 Page 2 — Industry Deep-Dive

- Matrix table of the highest-risk job roles within each industry (mirrors SQL query 8.2.2).
- Bubble/scatter chart plotting average routine-task percentage against high-risk percentage per industry, sized by employee count.

9.3 Page 3 — AI Adoption Analysis

- Bar chart of High Risk % by AI Adoption Level.
- Line chart of High Risk % against Number of AI Tools Used, showing a clear step-up pattern.
- Gauge visual showing the average Predicted_Risk_Score across all employees.
- Risk-tier breakdown table (Critical / Elevated / Stable), derived from the model's predicted risk score.

9.4 The Integration Point

What makes this more than a descriptive dashboard:

Page 3 surfaces each employee's out-of-sample Predicted_Risk_Score, generated by the cross-validated Random Forest model described in Section 7.6. This means the dashboard is not purely descriptive business intelligence — it is BI fed directly by a validated machine learning model's output, connecting all three tools used in this project into a single coherent deliverable.

9.5 Core DAX Measures

```
Total Employees = COUNTROWS(powerbi_dataset)
High Risk Count = SUM(powerbi_dataset[High_Risk_Flag])
High Risk % = DIVIDE([High Risk Count], [Total Employees], 0)
Avg Predicted Risk Score = AVERAGE(powerbi_dataset[Predicted_Risk_Score])
Model Accuracy % = DIVIDE(SUM(powerbi_dataset[Prediction_Match]),
```

```
[Total Employees], 0)
```


10. Key Business Insights

Synthesizing the EDA, modeling, and SQL findings, three insights stand out as the most actionable takeaways from this analysis.

10.1 Routine, Low-Creativity Roles Face Dramatically Higher Risk

Risk climbs from 0.3% to 77.1% as Routine_Task_Percentage moves from the lowest to the highest bucket (Section 5.4). This is the single clearest, most actionable signal in the entire dataset: any role audit aimed at identifying automation exposure should start by measuring routine-task content directly.

10.2 Entry-Level Employees Are the Most Exposed Group

At 44.5% high-risk rate versus 11.5% for Senior-level roles (Section 5.2.2), entry-level employees face roughly four times the exposure of senior employees. Organizations seeking to invest in workforce resilience should prioritize reskilling and upskilling programs at the entry tier first, where the protective effect of intervention is likely to be largest.

10.3 Correlation vs. Causation: A Necessary Caveat

Stated explicitly, not glossed over:

Higher individual AI tool usage correlates strongly with higher layoff risk (Section 5.2.3). It would be a mistake to interpret this as “using AI tools causes layoff risk.” The more plausible explanation is reverse: roles that are already highly automatable are the roles where organizations deploy the most AI tooling in the first place. The tool usage is a downstream consequence of role automatability, not an independent cause of risk. Distinguishing correlation from causation here is essential to drawing the right policy conclusion — the intervention point is the task structure of the role, not the employee's tool usage.

11. Limitations and Honest Reflection

No analysis is complete without an honest account of its limits. The following three limitations are disclosed unprompted, as part of the deliverable itself rather than as a defensive afterthought.

11.1 Likely Synthetic Data Generation

As discussed in Sections 3.4 and 7.4, the near-perfect separability of the risk classes along automation-related features suggests this dataset was built using an explicit generation rule rather than sampled from genuine organizational outcomes. The practical consequence is that this project's model performance numbers (92.4% out-of-sample accuracy, 0.981–0.996 ROC-AUC) demonstrate correct methodology — proper train/test splitting, class-imbalance handling, leakage-free cross-validation, and feature engineering — rather than constituting evidence that the model would perform comparably on real, organically collected workforce data.

11.2 No Temporal Dimension

The dataset represents a single snapshot in time, with no longitudinal or time-series structure. This means the analysis cannot show how layoff risk trends evolve as AI adoption increases within a given organization or role over time, nor can model predictions be validated against actual subsequent layoff events, since no such ground-truth outcome data exists beyond the labels already provided.

11.3 No Company-Level Financial Context

The dataset contains no company-level financial information — revenue trends, real historical layoff events, restructuring announcements, or profitability data — that would be necessary to validate the model's findings against actual organizational decision-making. Layoff decisions in practice are driven by a combination of financial pressure, strategic restructuring, and automation potential; this analysis addresses only the automation-potential dimension.

11.4 Why These Limitations Are Disclosed Here

Naming these limitations explicitly and unprompted is intended as a signal of analytical maturity rather than weakness. In a professional setting, the ability to recognize and articulate the boundaries of one's own analysis is at least as valuable as the analysis itself.

12. Conclusion

This project set out to demonstrate an end-to-end data analytics workflow spanning Python, SQL, and Power BI, applied to a concrete business question about AI-driven layoff risk. Across exploratory analysis, feature engineering, predictive modeling, structured querying, and dashboard design, the project surfaced a consistent and intuitive narrative: layoff risk concentrates predictably in routine, low-creativity, highly automatable roles, with entry-level employees bearing a disproportionate share of that exposure.

12.1 Skills Demonstrated

- End-to-end pipeline design: Python → SQL → Power BI, operating as one consistent workflow rather than three disconnected exercises.
- Exploratory data analysis and domain-informed feature engineering, including an engineered ratio feature that outperformed every raw column in the dataset.
- Rigorous model validation, including explicit avoidance of data leakage via cross-validated out-of-sample prediction.
- Business-focused SQL querying, including window functions, CTEs, and multi-metric summary tables designed for downstream BI consumption.
- Interactive dashboard design that integrates descriptive BI with validated machine learning output, rather than treating them as separate deliverables.

12.2 Closing Note

The author welcomes the opportunity to walk through any part of the underlying code, SQL queries, or dashboard in greater depth during a technical interview or portfolio review.